

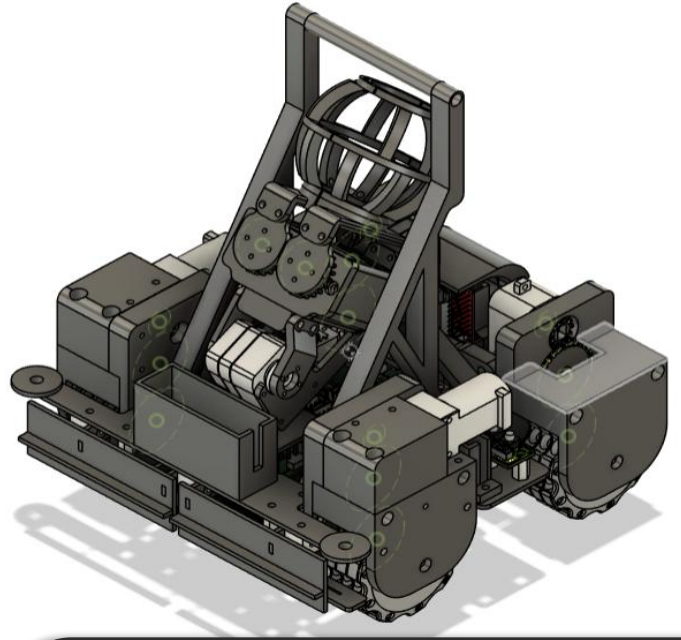


DMS-Kaede

Hardware: kouta Sakata

Electricity: kai Sonoda

Software: yakumo Kusanagi



Software

▪ Line Tracking

Controlled by five sensors and PD control. The central sensor detects derailment.

▪ Rescue

Use OpenMV. Detect silver → Move to the center → Rotate while searching for the ball → Find the zone corresponding to the color of the picked-up ball → Drop the ball → Repeat

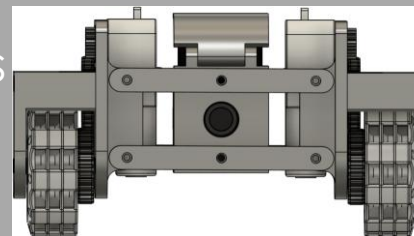
Mechanism

▪ Arm

There are three servo motors: two are used to move the base of the arm, and the other is for the arm that grasps the ball.

▪ Suspension

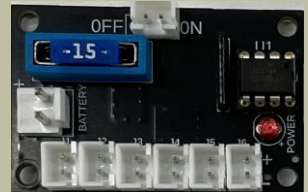
This design allows the motors on both sides to move up and down along the central support column, making it easier to navigate steps. Additionally, the support column is designed to accommodate the Open MV.



PCB

▪ BMS Circuit Board

When the LiFe battery is fully charged, the LED on the protection circuit stays lit; when the charge level drops below a certain point, it begins to flash.



▪ sensor

I'm using five sensors, all facing forward and lined up side by side.